AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Carbon monoxide (co)
Observed / expected baseline	0.4 / 0.143114 ppm
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-06-19 19:00 UTC
Location	29.6697, -95.1285
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	7 / 468	56.38 to 100; mean 83.8526	63.01 at 2026-06-19 18:55Z; 5 min before event
asos	Air temperature (temperature)	degC	7 / 468	23 to 35; mean 28.542	33 at 2026-06-19 18:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	7 / 467	0 to 340; mean 135.61	190 at 2026-06-19 18:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	7 / 468	0 to 8.74555; mean 3.4164	5.14444 at 2026-06-19 18:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	8 / 96	5867.19 to 5918.16; mean 5895.21	5910.07 at 2026-06-19 18:00Z; 1 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	8 / 96	64.9563 to 1814.28; mean 615.639	1635.88 at 2026-06-19 18:00Z; 1 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	8 / 96	40.017 to 59.8281; mean 50.5666	50.8807 at 2026-06-19 18:00Z; 1 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	8 / 96	1002.08 to 1012.77; mean 1008.14	1010.39 at 2026-06-19 18:00Z; 1 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	8 / 96	18.3317 to 23.2907; mean 20.2516	18.517 at 2026-06-19 18:00Z; 1 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	8 / 96	-3.27924 to 1.58572; mean -1.2137	0.269746 at 2026-06-19 18:00Z; 1 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	8 / 96	-1.84918 to 5.8484; mean 2.9406	4.38859 at 2026-06-19 18:00Z; 1 h before event
openaq	Ozone (ozone)	ppm	16 / 1066	-0.001 to 0.049; mean 0.02	0.031 at 2026-06-19 19:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 217	8 to 72; mean 34.4977	31 at 2026-06-19 19:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	79 / 2712	0 to 93.73; mean 13.7915	14.1 at 2026-06-19 19:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 361	0 to 100; mean 66.0637	95 at 2026-06-19 19:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 361	58 to 97; mean 84.6925	67 at 2026-06-19 19:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 361	0 to 3.93; mean 0.2092	0 at 2026-06-19 19:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 361	1003 to 1014; mean 1009.69	1012 at 2026-06-19 19:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 361	23.58 to 35.6; mean 28.8806	33.55 at 2026-06-19 19:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 361	0 to 360; mean 160.46	272 at 2026-06-19 19:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 361	0.45 to 5.42; mean 2.2325	1.34 at 2026-06-19 19:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	60 / 4263	0 to 359.8; mean 15.6561	6.7 at 2026-06-19 19:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-19 19:04Z; 5 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-19 19:04Z; 5 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	3 / 3	0.0278085 to 0.0281594; mean 0.0279	0.0278329 at 2026-06-19 19:04Z; 5 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-19 19:04Z; 5 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	3 / 3	9.81221e-05 to 0.000145702; mean 0.0001	0.000144456 at 2026-06-19 19:04Z; 5 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-19 19:04Z; 5 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	1 / 1	5.0006e-05 to 5.0006e-05; mean 0.0001	5.0006e-05 at 2026-06-19 19:31Z; 31 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	2 / 2	1 to 1; mean 1	1 at 2026-06-19 19:31Z; 31 min after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-19 19:04Z; 5 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	3 / 3	-4.97356e-05 to -4.29344e-06; mean -0	-8.26765e-06 at 2026-06-19 19:04Z; 5 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-19 19:04Z; 5 min after event
tceq	Carbon monoxide (co)	ppm	3 / 216	0.1 to 0.8; mean 0.2435	0.4 at 2026-06-19 19:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 814	-2.7 to 24.5; mean 5.1143	1.7 at 2026-06-19 19:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 209	-1.4 to 1.8; mean 0.0225	0.2 at 2026-06-19 19:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	06-18 06Z 95.09, 12Z 75.93, 18Z 66.52, 06-19 00Z 86.55, 06Z 93.22, 12Z 73.52, 18Z 71.84, 06-20 00Z 86.12, 06Z 91.18, 12Z 84.32, 18Z 92.09, 06-21 00Z 93.82, 06Z 96.57
asos	Air temperature (temperature)	06-18 06Z 26.51, 12Z 30.82, 18Z 33.24, 06-19 00Z 28.99, 06Z 27.4, 12Z 30.83, 18Z 31.42, 06-20 00Z 27.73, 06Z 27.13, 12Z 28.1, 18Z 24.4, 06-21 00Z 24.67, 06Z 25.06
asos	Wind direction (wind_direction)	06-18 06Z 150.6, 12Z 189.5, 18Z 172.9, 06-19 00Z 154.8, 06Z 139, 12Z 163.3, 18Z 157.6, 06-20 00Z 126.4, 06Z 126.9, 12Z 61.39, 18Z 125.6, 06-21 00Z 56.39, 06Z 21.67
asos	Wind speed (wind_speed)	06-18 06Z 2.146, 12Z 4.103, 18Z 5.255, 06-19 00Z 4.231, 06Z 3.137, 12Z 4.838, 18Z 3.865, 06-20 00Z 4.001, 06Z 2.687, 12Z 2.158, 18Z 3.058, 06-21 00Z 1.215, 06Z 0.5144
noaa_gfs	500-hPa geopotential height (gh_500)	06-18 12Z 5868, 18Z 5893, 06-19 00Z 5901, 06Z 5917, 12Z 5905, 18Z 5910, 06-20 00Z 5900, 06Z 5903, 12Z 5882, 18Z 5896, 06-21 00Z 5882, 06Z 5885
noaa_gfs	Planetary boundary layer height (pbl_height)	06-18 12Z 371.9, 18Z 1510, 06-19 00Z 742.9, 06Z 470.3, 12Z 523.5, 18Z 1568, 06-20 00Z 223.9, 06Z 535.7, 12Z 246.4, 18Z 846.9, 06-21 00Z 184.9, 06Z 162.5
noaa_gfs	Precipitable water (precipitable_water)	06-18 12Z 48.94, 18Z 45.81, 06-19 00Z 42.25, 06Z 41.08, 12Z 45.15, 18Z 51.53, 06-20 00Z 58.97, 06Z 54.86, 12Z 51.93, 18Z 55.93, 06-21 00Z 53.73, 06Z 56.61
noaa_gfs	Surface pressure (surface_pressure)	06-18 12Z 1003, 18Z 1004, 06-19 00Z 1004, 06Z 1007, 12Z 1009, 18Z 1010, 06-20 00Z 1009, 06Z 1011, 12Z 1011, 18Z 1011, 06-21 00Z 1008, 06Z 1009
noaa_gfs	850-hPa temperature (t_850)	06-18 12Z 19.78, 18Z 19.41, 06-19 00Z 22.14, 06Z 22.86, 12Z 21.86, 18Z 18.83, 06-20 00Z 19.98, 06Z 19.91, 12Z 18.5, 18Z 18.74, 06-21 00Z 21.48, 06Z 19.53
noaa_gfs	10-m eastward wind (u_10m)	06-18 12Z 0.3082, 18Z -0.3598, 06-19 00Z -1.914, 06Z -1.369, 12Z -0.8198, 18Z -0.5028, 06-20 00Z -1.788, 06Z -1.497, 12Z -0.8225, 18Z -2.864, 06-21 00Z -2.014, 06Z -0.921
noaa_gfs	10-m northward wind (v_10m)	06-18 12Z 2.847, 18Z 4.468, 06-19 00Z 4.332, 06Z 3.112, 12Z 3.448, 18Z 4.631, 06-20 00Z 2.991, 06Z 4.281, 12Z 2.435, 18Z 0.9756, 06-21 00Z -0.2617, 06Z 2.029
openaq	Ozone (ozone)	06-18 06Z 0.01596, 12Z 0.02766, 18Z 0.03645, 06-19 00Z 0.02774, 06Z 0.01758, 12Z 0.02026, 18Z 0.02624, 06-20 00Z 0.02021, 06Z 0.01288, 12Z 0.01231, 18Z 0.0135, 06-21 00Z 0.008507, 06Z 0.002611
openaq	PM10 (pm10)	06-18 06Z 23.33, 12Z 51.11, 18Z 49.29, 06-19 00Z 44.67, 06Z 38.71, 12Z 48.44, 18Z 45.94, 06-20 00Z 23.06, 06Z 30.39, 12Z 30.44, 18Z 11.61, 06-21 00Z 19.28, 06Z 25.17
openaq	PM2.5 (pm25)	06-18 06Z 10.36, 12Z 30.26, 18Z 20.34, 06-19 00Z 33.57, 06Z 21.73, 12Z 14.42, 18Z 10.18, 06-20 00Z 6.631, 06Z 5.896, 12Z 5.46, 18Z 3.105, 06-21 00Z 4.181, 06Z 5.035
openweather	Cloud cover (cloud_cover)	06-18 06Z 2.346, 12Z 39.5, 18Z 19.4, 06-19 00Z 6.931, 06Z 80.77, 12Z 72.39, 18Z 68.6, 06-20 00Z 97.41, 06Z 94.23, 12Z 99.5, 18Z 99.61, 06-21 00Z 98.57, 06Z 82.6
openweather	Relative humidity (humidity)	06-18 06Z 93.35, 12Z 82.5, 18Z 70.27, 06-19 00Z 83.59, 06Z 91.3, 12Z 81.81, 18Z 72.73, 06-20 00Z 86.86, 06Z 91.13, 12Z 86.37, 18Z 82.23, 06-21 00Z 93.8, 06Z 95.2
openweather	Precipitation rate (precipitation)	06-18 06Z 0, 12Z 0.009, 18Z 0, 06-19 00Z 0, 06Z 0, 12Z 0, 18Z 0.035, 06-20 00Z 0.1752, 06Z 0.43, 12Z 1.097, 18Z 0.7358, 06-21 00Z 0.01667, 06Z 0
openweather	Surface pressure (pressure)	06-18 06Z 1004, 12Z 1006, 18Z 1005, 06-19 00Z 1008, 06Z 1009, 12Z 1011, 18Z 1011, 06-20 00Z 1012, 06Z 1012, 12Z 1014, 18Z 1012, 06-21 00Z 1011, 06Z 1011
openweather	Air temperature (temperature)	06-18 06Z 26.13, 12Z 30.23, 18Z 34.37, 06-19 00Z 29.97, 06Z 27.74, 12Z 30.18, 18Z 32.75, 06-20 00Z 27.75, 06Z 27.41, 12Z 28.55, 18Z 27.55, 06-21 00Z 24.3, 06Z 24.4
openweather	Wind direction (wind_direction)	06-18 06Z 199.5, 12Z 177.4, 18Z 178.3, 06-19 00Z 158.7, 06Z 165.5, 12Z 163.5, 18Z 185.2, 06-20 00Z 135.2, 06Z 164.8, 12Z 115.8, 18Z 161.4, 06-21 00Z 121.6, 06Z 177.6
openweather	Wind speed (wind_speed)	06-18 06Z 2.185, 12Z 2.35, 18Z 2.545, 06-19 00Z 2.314, 06Z 2.298, 12Z 2.28, 18Z 2.305, 06-20 00Z 1.742, 06Z 2.456, 12Z 1.399, 18Z 3.311, 06-21 00Z 1.668, 06Z 1.508

Source	Metric	Values
purpleair	PM2.5 (pm25)	06-18 06Z 19.35, 12Z 30.5, 18Z 24.65, 06-19 00Z 33.68, 06Z 24.1, 12Z 15.23, 18Z 11.65, 06-20 00Z 7.268, 06Z 6.837, 12Z 6.451, 18Z 5.188, 06-21 00Z 6.047, 06Z 8.052
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	06-18 18Z 1, 06-19 18Z 1, 06-20 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	06-18 18Z 1, 06-19 18Z 1, 06-20 18Z 1
sentinel5p	CO column (s5p_co_column)	06-18 18Z 0.02816, 06-19 18Z 0.02782
sentinel5p	CO granules available (s5p_co_granule_available)	06-18 18Z 1, 06-19 18Z 1, 06-20 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	06-18 18Z 9.812e-05, 06-19 18Z 0.0001451
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	06-18 18Z 1, 06-19 18Z 1, 06-20 18Z 1
sentinel5p	NO2 granules available (s5p_no2_granule_available)	06-19 18Z 1, 06-20 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	06-18 18Z 1, 06-19 18Z 1, 06-20 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	06-18 18Z -4.293e-06, 06-19 18Z -2.9e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	06-18 18Z 1, 06-19 18Z 1, 06-20 18Z 1
tceq	Carbon monoxide (co)	06-18 06Z 0.26, 12Z 0.2933, 18Z 0.2278, 06-19 00Z 0.1944, 06Z 0.1944, 12Z 0.2278, 18Z 0.2222, 06-20 00Z 0.2833, 06Z 0.1667, 12Z 0.2, 18Z 0.2944, 06-21 00Z 0.3556, 06Z 0.2833
tceq	Nitrogen dioxide (no2)	06-18 06Z 5.483, 12Z 4.287, 18Z 2.812, 06-19 00Z 3.226, 06Z 3.9, 12Z 4.366, 18Z 4.846, 06-20 00Z 7.707, 06Z 2.27, 12Z 4.406, 18Z 6.483, 06-21 00Z 10.06, 06Z 8.754
tceq	Sulfur dioxide (so2)	06-18 06Z -0.05333, 12Z 0.1467, 18Z 0.1944, 06-19 00Z 0.4611, 06Z -0.05556, 12Z 0.06111, 18Z 0.005556, 06-20 00Z 0.01111, 06Z -0.1, 12Z -0.06111, 18Z -0.1, 06-21 00Z -0.1273, 06Z -0.4667

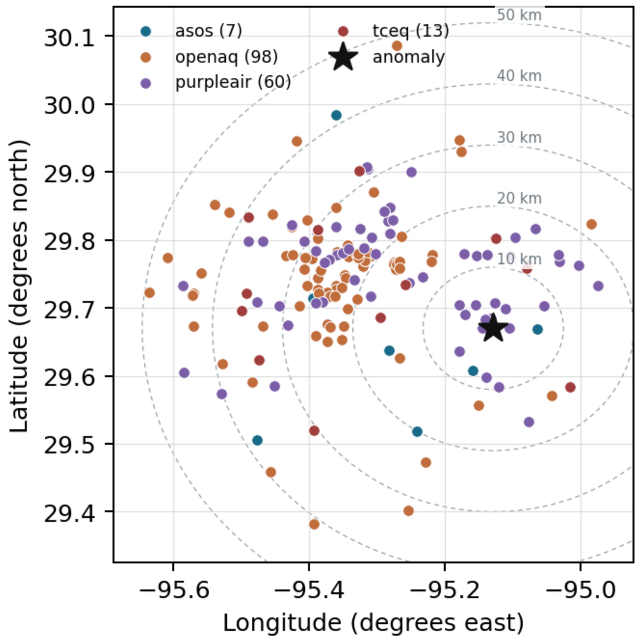
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

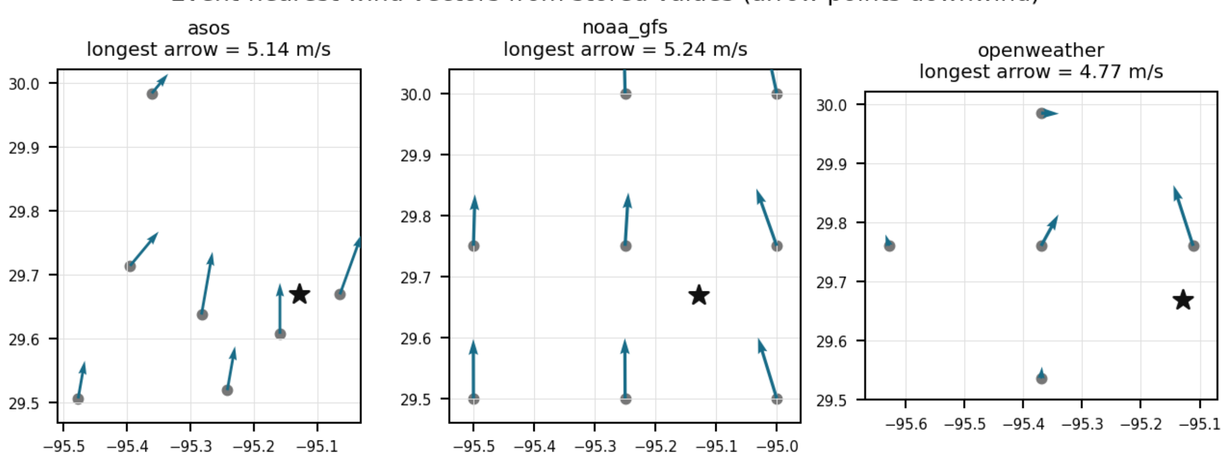
Source	Metric	Values
asos	Relative humidity (humidity)	T41 (6.225 km) 84.46, EFD (7.53 km) 80.27, HOU (15.295 km) 83.67, LVJ (20.026 km) 83.98, MCJ (26.207 km) 86.18, AXH (38.286 km) 80.87, IAH (41.543 km) 85.75
asos	Air temperature (temperature)	T41 (6.225 km) 28.64, EFD (7.53 km) 30.22, HOU (15.295 km) 28.67, LVJ (20.026 km) 28.5, MCJ (26.207 km) 28.19, AXH (38.286 km) 28.5, IAH (41.543 km) 27.91
asos	Wind direction (wind_direction)	T41 (6.225 km) 153.8, EFD (7.53 km) 158.6, HOU (15.295 km) 144, LVJ (20.026 km) 122.5, MCJ (26.207 km) 150.6, AXH (38.286 km) 106.4, IAH (41.543 km) 125.1
asos	Wind speed (wind_speed)	T41 (6.225 km) 4.13, EFD (7.53 km) 4.873, HOU (15.295 km) 3.794, LVJ (20.026 km) 2.737, MCJ (26.207 km) 4.587, AXH (38.286 km) 1.701, IAH (41.543 km) 2.822
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (14.741 km) 5895, gfs:29.75,-95.00 (15.288 km) 5895, gfs:29.50,-95.25 (22.231 km) 5896, gfs:29.50,-95.00 (22.598 km) 5896, gfs:29.75,-95.50 (36.97 km) 5895, gfs:30.00,-95.25 (38.548 km) 5894, gfs:30.00,-95.00 (38.76 km) 5894, gfs:29.50,-95.50 (40.577 km) 5896
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (14.741 km) 662.9, gfs:29.75,-95.00 (15.288 km) 589, gfs:29.50,-95.25 (22.231 km) 556.9, gfs:29.50,-95.00 (22.598 km) 627.2, gfs:29.75,-95.50 (36.97 km) 734.7, gfs:30.00,-95.25 (38.548 km) 679.5, gfs:30.00,-95.00 (38.76 km) 500, gfs:29.50,-95.50 (40.577 km) 575
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (14.741 km) 50.72, gfs:29.75,-95.00 (15.288 km) 51, gfs:29.50,-95.25 (22.231 km) 50.42, gfs:29.50,-95.00 (22.598 km) 50.56, gfs:29.75,-95.50 (36.97 km) 50.31, gfs:30.00,-95.25 (38.548 km) 50.29, gfs:30.00,-95.00 (38.76 km) 50.93, gfs:29.50,-95.50 (40.577 km) 50.3

Source	Metric	Values
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (14.741 km) 1008, gfs:29.75,-95.00 (15.288 km) 1009, gfs:29.50,-95.25 (22.231 km) 1009, gfs:29.50,-95.00 (22.598 km) 1010, gfs:29.75,-95.50 (36.97 km) 1007, gfs:30.00,-95.25 (38.548 km) 1007, gfs:30.00,-95.00 (38.76 km) 1008, gfs:29.50,-95.50 (40.577 km) 1008
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (14.741 km) 20.17, gfs:29.75,-95.00 (15.288 km) 20.29, gfs:29.50,-95.25 (22.231 km) 20.28, gfs:29.50,-95.00 (22.598 km) 20.47, gfs:29.75,-95.50 (36.97 km) 20.18, gfs:30.00,-95.25 (38.548 km) 20.16, gfs:30.00,-95.00 (38.76 km) 20.13, gfs:29.50,-95.50 (40.577 km) 20.33
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (14.741 km) -0.9586, gfs:29.75,-95.00 (15.288 km) -1.229, gfs:29.50,-95.25 (22.231 km) -1.194, gfs:29.50,-95.00 (22.598 km) -1.235, gfs:29.75,-95.50 (36.97 km) -1.134, gfs:30.00,-95.25 (38.548 km) -1.226, gfs:30.00,-95.00 (38.76 km) -1.352, gfs:29.50,-95.50 (40.577 km) -1.381
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (14.741 km) 2.649, gfs:29.75,-95.00 (15.288 km) 2.96, gfs:29.50,-95.25 (22.231 km) 3.291, gfs:29.50,-95.00 (22.598 km) 3.899, gfs:29.75,-95.50 (36.97 km) 2.717, gfs:30.00,-95.25 (38.548 km) 2.361, gfs:30.00,-95.00 (38.76 km) 2.676, gfs:29.50,-95.50 (40.577 km) 2.972
openaq	Ozone (ozone)	332 (0.032 km) 0.02426, 2800 (11.598 km) 0.01896, 2039 (14.096 km) 0.01774, 320 (14.272 km) 0.01632, 3378 (14.356 km) 0.01607, 5077791 (14.558 km) 0.017, 3214 (14.765 km) 0.0214, 325 (16.163 km) 0.02185, +8 more
openaq	PM10 (pm10)	271 (0.032 km) 28.01, 13380082 (14.356 km) 34.86, 15852419 (21.358 km) 40.79
openaq	PM2.5 (pm25)	268 (0.032 km) 14, 13955815 (12.727 km) 10.28, 14404954 (13.725 km) 14.95, 14157975 (14.275 km) 12.55, 3379 (14.356 km) 12.3, 15539682 (14.359 km) 11.66, 8967123 (14.413 km) 13.47, 8967479 (14.413 km) 11.69, +71 more
openweather	Cloud cover (cloud_cover)	grid:east (10.221 km) 66.67, grid:center (25.388 km) 66.05, grid:south (27.676 km) 60.61, grid:north (42.079 km) 72.71, grid:west (49.325 km) 64.28
openweather	Relative humidity (humidity)	grid:east (10.221 km) 86.61, grid:center (25.388 km) 83.95, grid:south (27.676 km) 83.86, grid:north (42.079 km) 85.11, grid:west (49.325 km) 83.94
openweather	Precipitation rate (precipitation)	grid:east (10.221 km) 0.119, grid:center (25.388 km) 0.1788, grid:south (27.676 km) 0.1361, grid:north (42.079 km) 0.3194, grid:west (49.325 km) 0.2929
openweather	Surface pressure (pressure)	grid:east (10.221 km) 1010, grid:center (25.388 km) 1010, grid:south (27.676 km) 1010, grid:north (42.079 km) 1010, grid:west (49.325 km) 1010
openweather	Air temperature (temperature)	grid:east (10.221 km) 28.94, grid:center (25.388 km) 29, grid:south (27.676 km) 29.08, grid:north (42.079 km) 28.53, grid:west (49.325 km) 28.85
openweather	Wind direction (wind_direction)	grid:east (10.221 km) 163.1, grid:center (25.388 km) 161.8, grid:south (27.676 km) 175.3, grid:north (42.079 km) 145.3, grid:west (49.325 km) 156.8
openweather	Wind speed (wind_speed)	grid:east (10.221 km) 3.43, grid:center (25.388 km) 2.22, grid:south (27.676 km) 1.841, grid:north (42.079 km) 2.091, grid:west (49.325 km) 1.582
purpleair	PM2.5 (pm25)	128007 (1.625 km) 11.08, 161689 (1.727 km) 15.29, 222589 (1.859 km) 15.54, 268165 (2.296 km) 17.73, 222593 (3.655 km) 15.66, 250205 (4.119 km) 12.8, 98633 (4.643 km) 8.932, 244939 (4.684 km) 10.52, +52 more
tceq	Carbon monoxide (co)	482011039 (0.0 km) 0.1466, 482011035 (14.355 km) 0.1371, 482011052 (29.756 km) 0.4425
tceq	Nitrogen dioxide (no2)	482011039 (0.0 km) 3.828, 482011015 (10.998 km) 5.186, 482011035 (14.355 km) 7.424, 482011050 (14.566 km) 1.097, 482010026 (14.787 km) 7.54, 482010416 (16.162 km) 2.885, 482011052 (29.756 km) 14.22, 480391004 (30.447 km) 1.682, +4 more
tceq	Sulfur dioxide (so2)	482011039 (0.0 km) 0.24, 482011035 (14.355 km) -0.3765, 482010051 (33.805 km) 0.1901

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

Localized industrial or heavy combustion activities near monitor 482011039 generated a localized elevation in ground-level carbon monoxide reaching 0.4 ppm at 19:00 UTC on June 19, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 36

A carbon monoxide concentration of 0.4 ppm was recorded at TCEQ monitor 482011039 in Houston at 19:00 UTC on June 19, 2026, representing a severe statistical anomaly relative to expected baseline values.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 36

Daytime boundary layer heights exceeding 1500 meters rapidly diluted the plume regionally, constraining the severe carbon monoxide footprint to the immediate vicinity of the local TCEQ monitor.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 36

The CO, SO2, and O3 levels were also elevated during the air quality anomaly, indicating potential contributions from industrial activities and vehicle emissions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 36

ASOS weather station reports and GFS atmospheric models indicated moderate south-southeasterly surface winds around 3.8 to 5.1 m/s and a daytime planetary boundary layer height exceeding 1500 meters during the anomaly period.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 36

The elevated carbon monoxide measurement at 19:00 UTC represented a z-score of approximately 5.19 above the expected value.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 36

Sentinel-5P satellite observations near the anomaly site recorded a carbon monoxide total column density of approximately 0.0278 mol/m^2 at 19:04 UTC on June 19, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 36

A transient low-level transport or fumigation contribution is supported because surface winds near the anomaly time were predominantly from the south to south-southeast at about 3.9 to 5.1 m/s while the boundary layer was deep at about 1636 m, which would permit advection of an upwind plume to the monitor without requiring strong local stagnation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 36

The anomaly occurred on June 19th at approximately 19:01:49 UTC, with the nearest sensor located 1.625 km away.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 36

The tceq carbon monoxide anomaly occurred at monitor 482011039, which has a mean carbon monoxide concentration of 0.1466 ppm over the provided context window and was the nearest monitor to the anomaly location at 0.0 km.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 36

South to south-southeasterly low-level flow likely transported emissions toward the anomaly site at the event time, because nearby surface winds were predominantly from about 158 to 190 degrees with moderate speeds around 3.9 to 5.1 m/s, placing emission sources located to the south or south-southeast of the monitor in the most plausible upwind sector.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 1.34 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.3003492113671715 m/s; n=361 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 36

The temperature and wind direction data suggest that the air quality anomaly was influenced by a heatwave event with stagnant atmospheric conditions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 36

South-southeasterly surface winds averaging 3.8 m/s to 5.1 m/s transported upwind industrial or traffic emissions directly toward the monitoring location.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 36

The tceq carbon monoxide monitor at 29.670, -95.129 measured 0.4 ppm at 2026-06-19T19:00:00+00:00, compared with an expected value of 0.1431137724550898 ppm, corresponding to a z-score of 5.187155188286583 and a severe anomaly classification.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 36

A nearby primary combustion source such as on-road traffic, diesel equipment, marine activity, rail operations, or industrial flaring is the leading physical cause of the Houston CO spike because the CO increase at the anomaly monitor was large while co-located NO_2 and SO_2 remained low and regional particulate levels were not concurrently elevated, which is consistent with a localized CO-rich plume rather than a broad secondary pollution event.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 36

Sentinel-5P satellite observations indicated standard regional total column carbon monoxide values around 0.0278 mol/m^2, confirming the surface anomaly was a localized ground-level event.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 36

A severe carbon monoxide anomaly was recorded on June 19, 2026, at 19:00 UTC at coordinates (29.670, -95.129) with a concentration of 0.4 ppm compared to an expected baseline of 0.143 ppm.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 36

A localized primary combustion plume is supported because the anomaly-site CO reached 0.4 ppm while co-located NO_2 was only 1.7 ppb and co-located SO_2 was 0.2 ppb, which is more consistent with a nearby CO-rich source than with a broad urban traffic or sulfur-bearing plume at the monitor.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 36

Regional observations do not support a broad combustion or sulfur-rich event because Sentinel-5P carbon monoxide column near the anomaly time was about 0.0278 mol/m^2 without enhancement, sulfur dioxide column was near zero or negative, and nearby particulate matter and ozone were not unusually high at the anomaly site.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 36

TCEQ ambient air monitoring data showed minimal concurrent sulfur dioxide levels (0.2 ppb) and low nitrogen dioxide concentrations (1.7 ppb) at the site during the carbon monoxide peak.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 36

The air quality anomaly is related to particulate matter (PM2.5) with a value of 6.7 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 36

The Houston TCEQ monitor 482011039 recorded a severe local carbon monoxide spike of 0.4 ppm at 2026-06-19T19:00:00+00:00 against an expected value of 0.143 ppm, while the same monitor reported only 1.7 ppb nitrogen dioxide and 0.2 ppb sulfur dioxide at that time.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 36

At 2026-06-19T19:00:00+00:00 at the same location, tceq nitrogen dioxide was 1.7 ppb and tceq sulfur dioxide was 0.2 ppb, while openaq ozone at 0.032 km was 0.031 ppm.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 36

A severe thunderstorm or tornado event occurred in the area, stirring up pollutants and reducing air quality.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 36

The air quality anomaly in Houston, Texas is likely caused by a combination of factors including high levels of PM2.5 and NO2 emissions from industrial activities and vehicle traffic.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 36

The anomaly is a result of a combination of factors including high temperatures, humidity, and stagnant atmospheric conditions, leading to poor air mixing and increased pollutant concentrations.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 36

A regional combustion or major petrochemical upset is contradicted because Sentinel-5P showed no elevated CO column over the area, satellite SO2 remained near zero, and nearby particulate matter was modest to low around the anomaly time rather than strongly elevated as in a large-area smoke or flaring episode.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 36

Ground-level carbon monoxide observations from TCEQ monitor 482011039 recorded a spike to 0.4 ppm at 19:00 UTC on June 19, 2026, whereas regional Sentinel-5P total carbon monoxide column concentrations showed no corresponding broad elevation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 36

Houston-area winds near 2026-06-19T19:00:00+00:00 were predominantly from the south to south-southeast with moderate speed, and modeled boundary-layer height was high near 1636 m, which is consistent with transport and daytime fumigation of a nearby plume rather than stagnant trapping.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 36

Surface meteorological observations during the anomaly period recorded south-southeasterly winds at speeds between 3.8 and 5.1 m/s and a daytime planetary boundary layer height exceeding 1500 meters.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 36

The anomaly is not related to any other pollutants such as CO, NO2, SO2, or O3, as their values are within normal ranges.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 36

Active hydrocarbon emissions in the Houston industrial corridor are a plausible contributing factor because formaldehyde columns were elevated on the same afternoon while satellite CO columns were not regionally enhanced, which is more consistent with localized petrochemical VOC activity or oxidation occurring alongside a surface CO plume than with a metropolitan-scale combustion buildup.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 36 of 36

The air quality anomaly in Houston, Texas is caused by high levels of SO_2 particles.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.